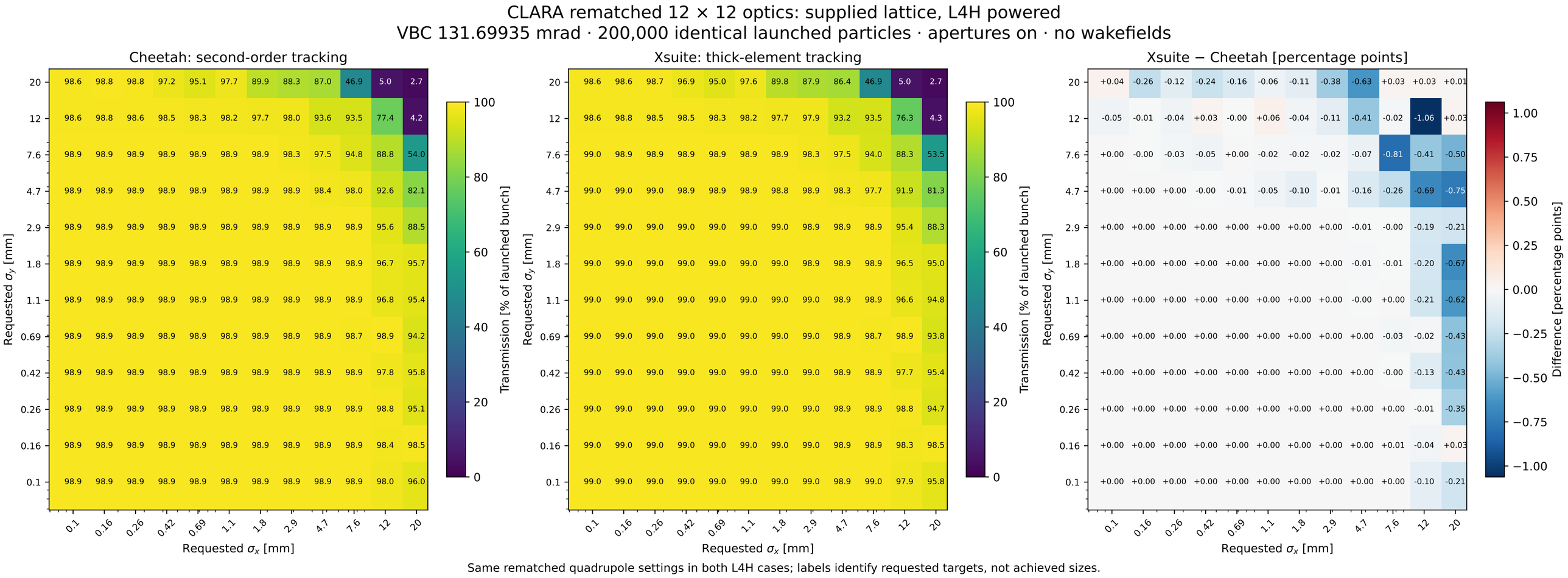
CLARA rematched 12 × 12 optics: supplied lattice, L4H powered
VBC 131.69935 mrad · 200,000 identical launched particles · apertures on · no wakefields
Cheetah: second-order tracking
Xsuite: thick-element tracking
Xsuite − Cheetah [percentage points]
Requested σ_y [mm]
Requested σ_x [mm]
Transmission [% of launched bunch]
Difference [percentage points]
20 12 7.6 4.7 2.9 1.8 1.1 0.69 0.42 0.26 0.16 0.1
0.1 0.16 0.26 0.42 0.69 1.1 1.8 2.9 4.7 7.6 12 20
0 20 40 60 80 100
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
98.6 98.8 98.8 97.2 95.1 97.7 89.9 88.3 87.0 46.9 5.0 2.7
98.6 98.8 98.6 98.5 98.3 98.2 97.7 98.0 93.6 93.5 77.4 4.2
98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.3 97.5 94.8 88.8 54.0
98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.4 98.0 92.6 82.1
98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 95.6 88.5
98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 96.7 95.7
98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 96.8 95.4
98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.7 98.9 94.2
98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 97.8 95.8
98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.8 95.1
98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.4 98.5
98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.9 98.0 96.0
98.6 98.6 98.7 96.9 95.0 97.6 89.8 87.9 86.4 46.9 5.0 2.7
98.6 98.8 98.5 98.5 98.3 98.2 97.7 97.9 93.2 93.5 76.3 4.3
99.0 98.9 98.9 98.9 98.9 98.9 98.9 98.3 97.5 94.0 88.3 53.5
99.0 99.0 99.0 98.9 98.9 98.9 98.8 98.9 98.3 97.7 91.9 81.3
99.0 99.0 99.0 99.0 99.0 99.0 99.0 98.9 98.9 98.9 95.4 88.3
99.0 99.0 99.0 99.0 99.0 99.0 99.0 98.9 98.9 98.9 96.5 95.0
99.0 99.0 99.0 99.0 99.0 99.0 99.0 99.0 98.9 98.9 96.6 94.8
99.0 99.0 99.0 99.0 99.0 99.0 99.0 99.0 98.9 98.7 98.9 93.8
99.0 99.0 99.0 99.0 99.0 99.0 99.0 99.0 98.9 98.9 97.7 95.4
99.0 99.0 99.0 99.0 99.0 99.0 99.0 99.0 98.9 98.9 98.8 94.7
99.0 99.0 99.0 99.0 99.0 99.0 99.0 99.0 98.9 98.9 98.3 98.5
99.0 99.0 99.0 99.0 99.0 99.0 99.0 99.0 98.9 99.0 97.9 95.8
+0.04 -0.26 -0.12 -0.24 -0.16 -0.06 -0.11 -0.38 -0.63 +0.03 +0.03 +0.01
-0.05 -0.01 -0.04 +0.03 -0.00 +0.06 -0.04 -0.11 -0.41 -0.02 -1.06 +0.03
+0.00 -0.00 -0.03 -0.05 +0.00 -0.02 -0.02 -0.02 -0.07 -0.81 -0.41 -0.50
+0.00 +0.00 +0.00 -0.00 -0.01 -0.05 -0.10 -0.01 -0.16 -0.26 -0.69 -0.75
+0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 -0.01 -0.00 -0.19 -0.21
+0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 -0.01 -0.01 -0.20 -0.67
+0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 -0.00 +0.00 -0.21 -0.62
+0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 -0.03 -0.02 -0.43
+0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 -0.00 -0.13 -0.43
+0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 -0.01 -0.35
+0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.01 -0.04 +0.03
+0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00 -0.10 -0.21
Same rematched quadrupole settings in both L4H cases; labels identify requested targets, not achieved sizes.